Department of Electrical, Computer, & Biomedical Engineering

Course Number	ELE734
Course Title	Low Power Digital Integrated Circuits
Semester/Year	F2025
Instructor	Andy Ye
TA Name	Esraa Alaa Aldeen

Lab/Tutorial Report No.	1

Report Title	CMOS Inverter

Section No.	5
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Due Date	November 4, 2025

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*By signing above you attest that you have contributed to this written lab report and confirm that all work you have contributed to this lab report is your own work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a "0" on the work, an "F" in the course, or possibly more severe penalties, as well as a Disciplinary Notice on your academic record under the Student Code of Academic Conduct, which can be found online at: www.ryerson.ca/senate/current/pol60.pdf.

1.1 Schematic of CMOS Inverter:

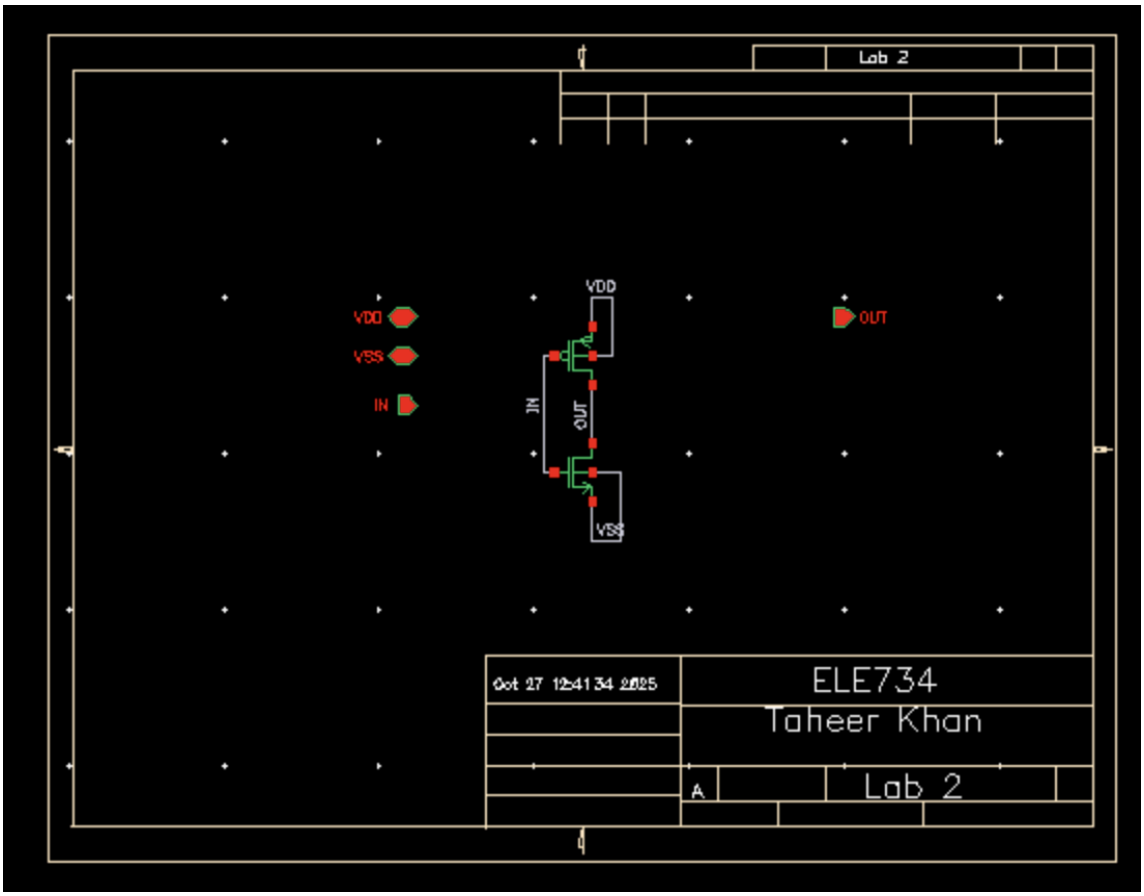


Figure 1: Schematic for CMOS Inverter:

1.2 Schematic Of Testbench:

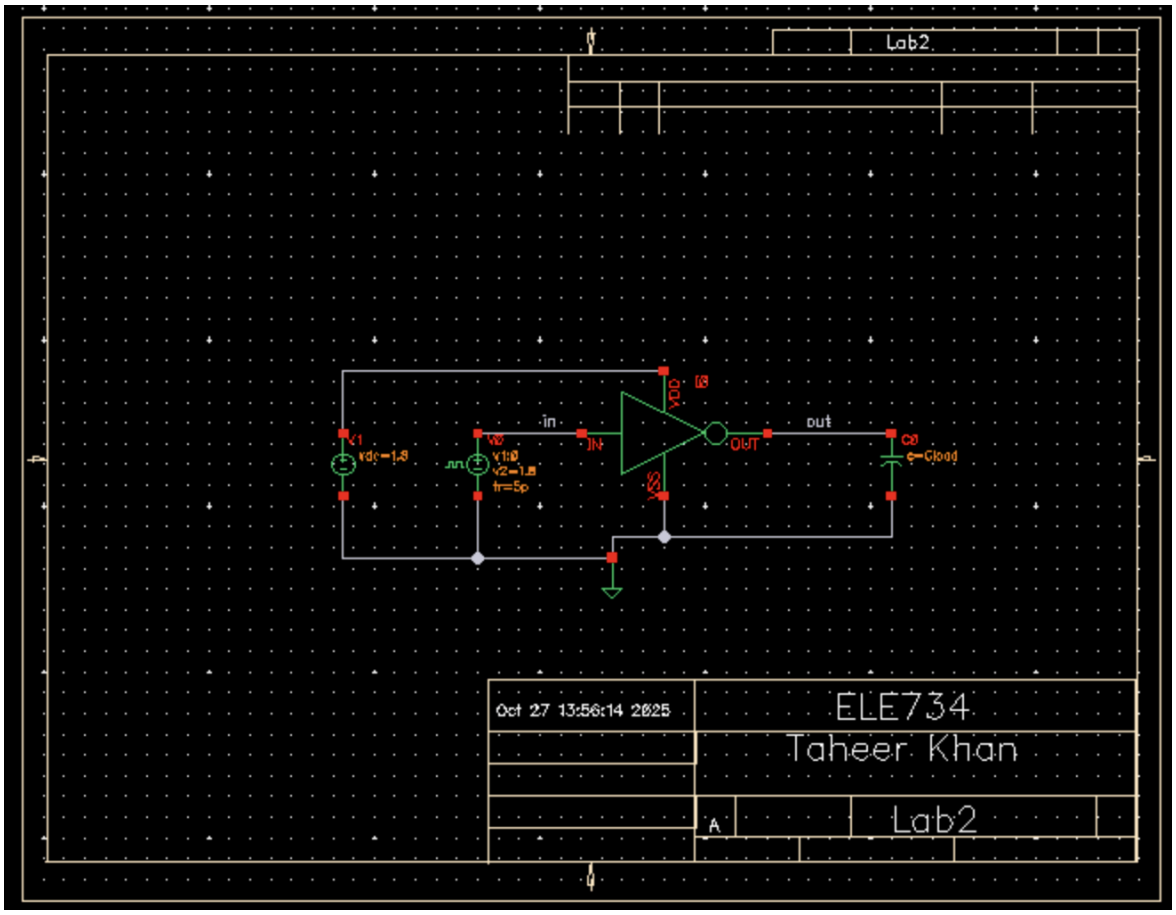
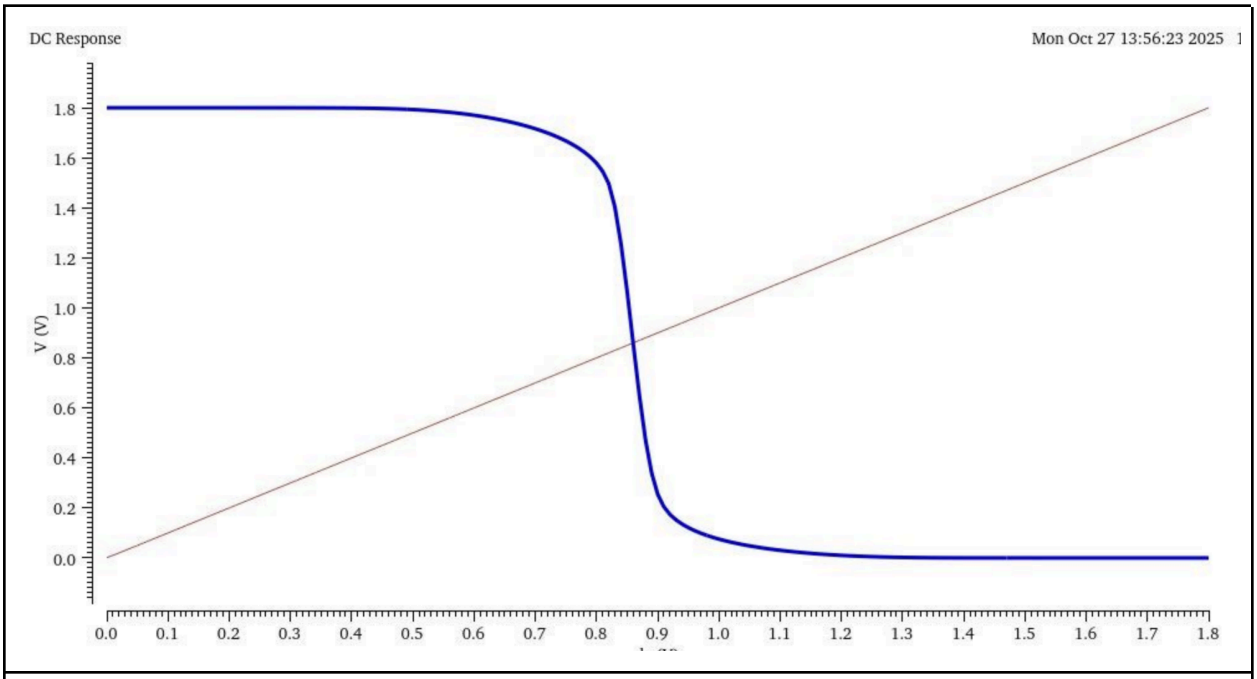


Figure 2: Schematic of Testbench of CMOS Inverter.

1.3 Result of Static & Dynamic CMOS Inverter Simulations:



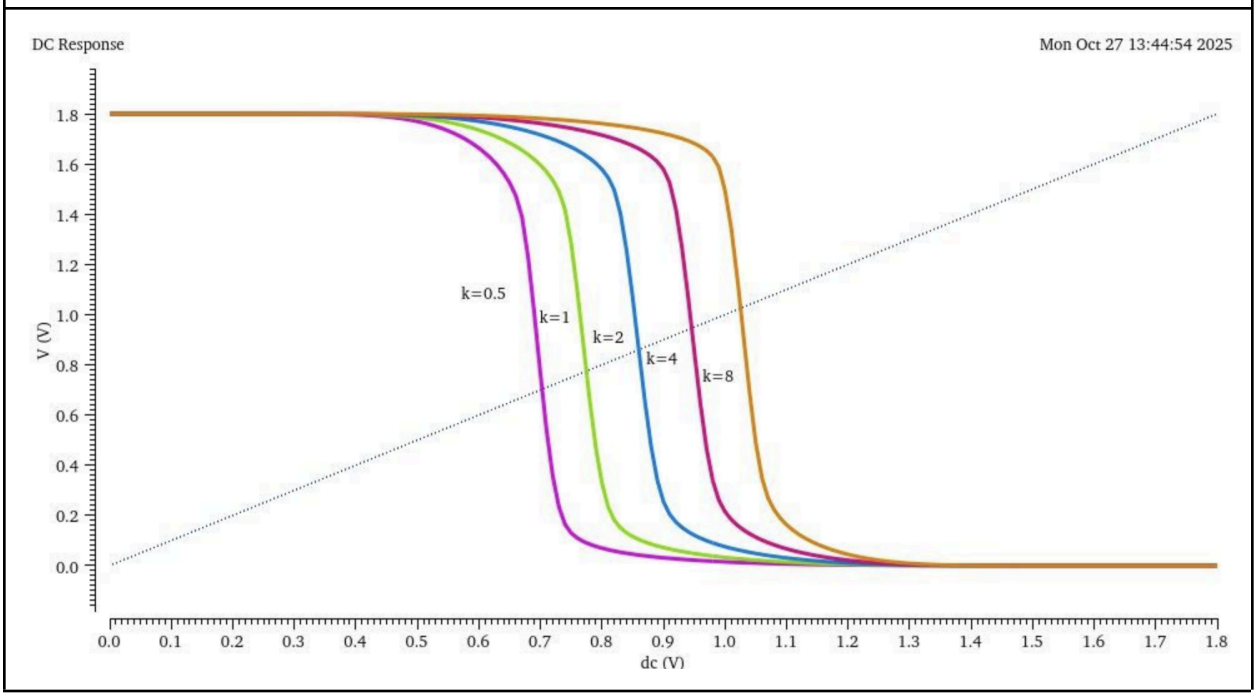


Table 1: CMOS DC Characteristics.

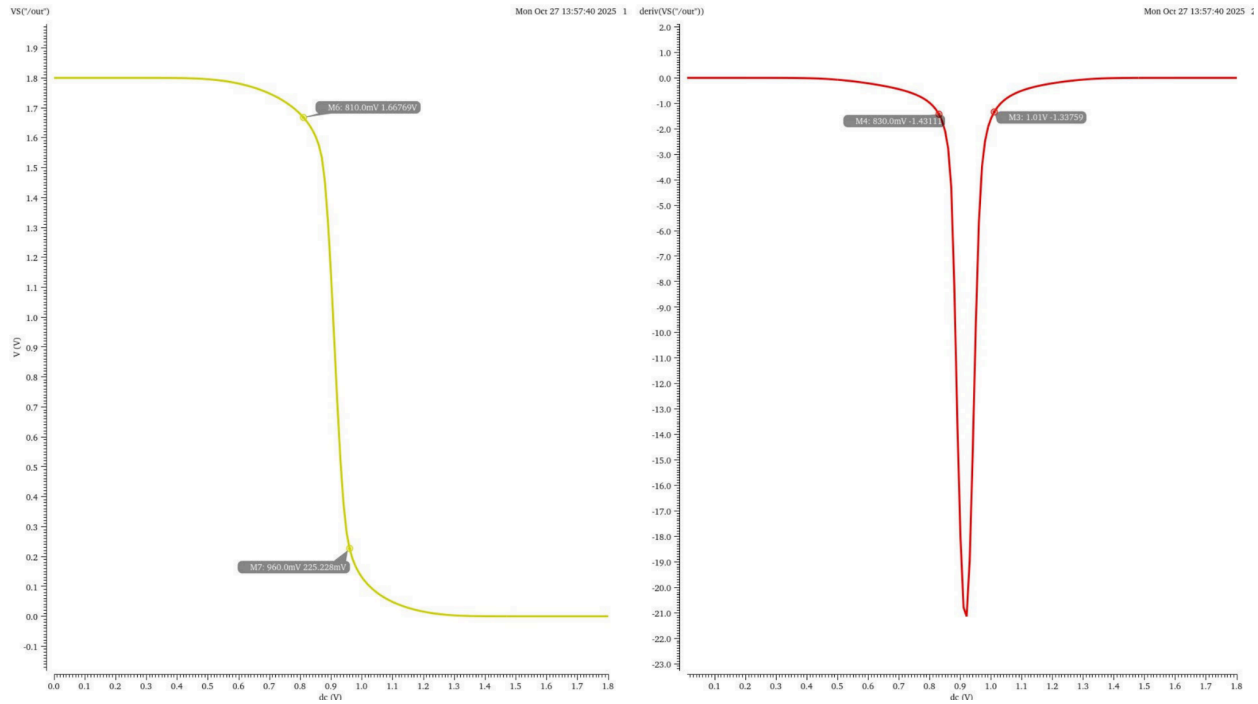


Figure 3: Derivative of CMOS Inverter DC Characteristics.

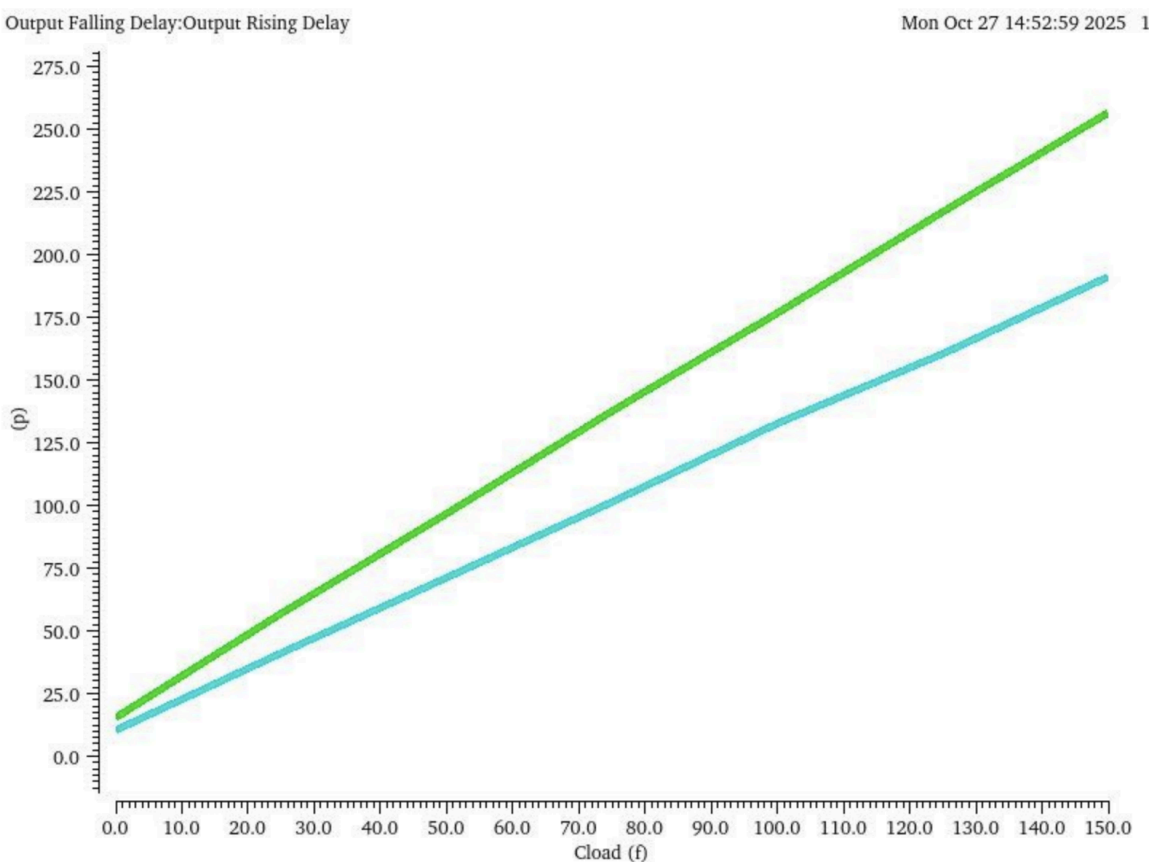


Figure 4: CMOS Inverter Rising and Falling Delay as a Function of Load Capacitance.

Output Delay Measurements

Rising Delay (s)	73.56p
Falling Delay (s)	96.34p

Table 2: Output Y Delay Measurements.

1.4 Layout Of CMOS Inverter:

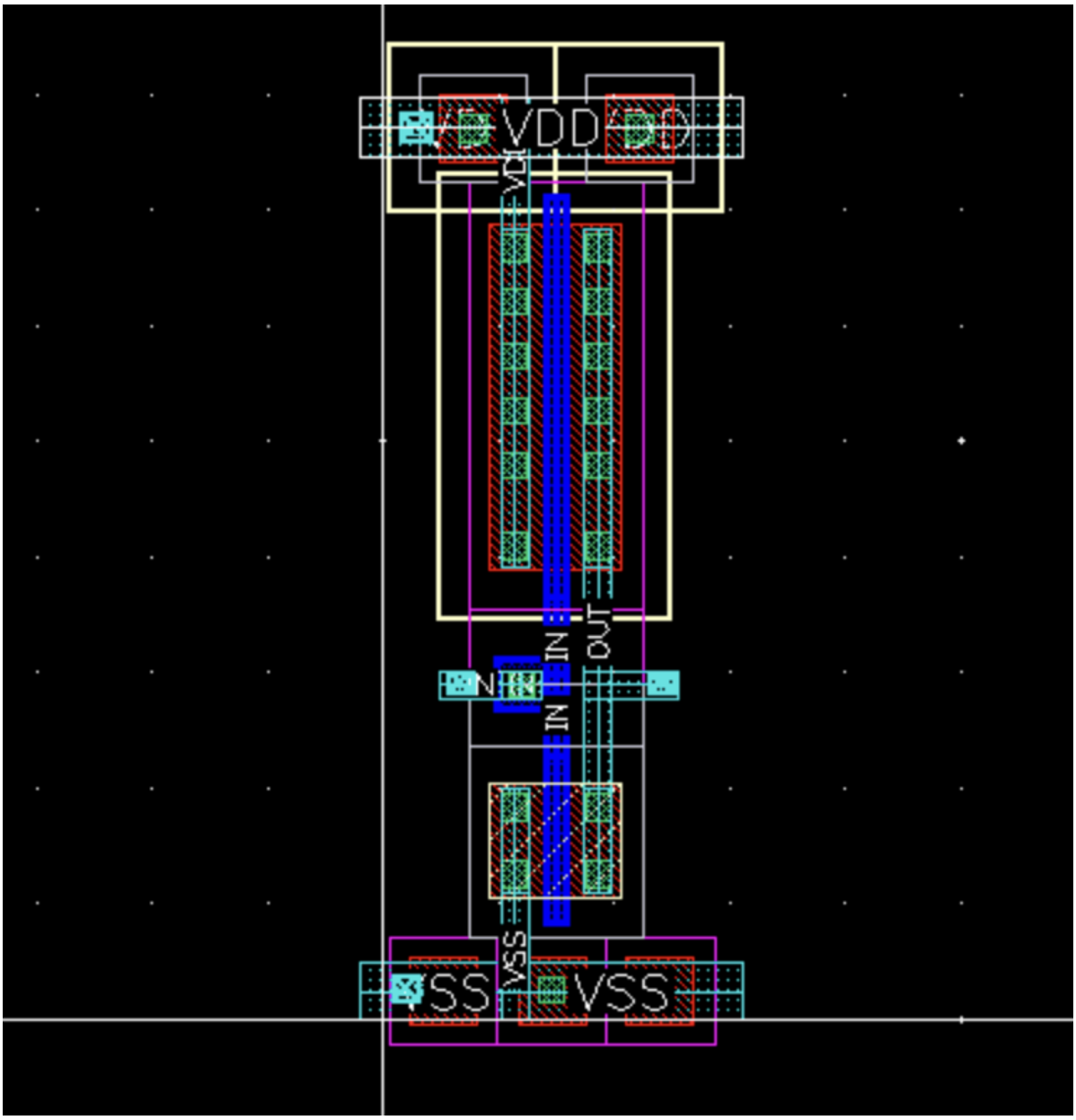


Table 4: Layout of CMOS Inverter

1.5 Post Layout Simulations:

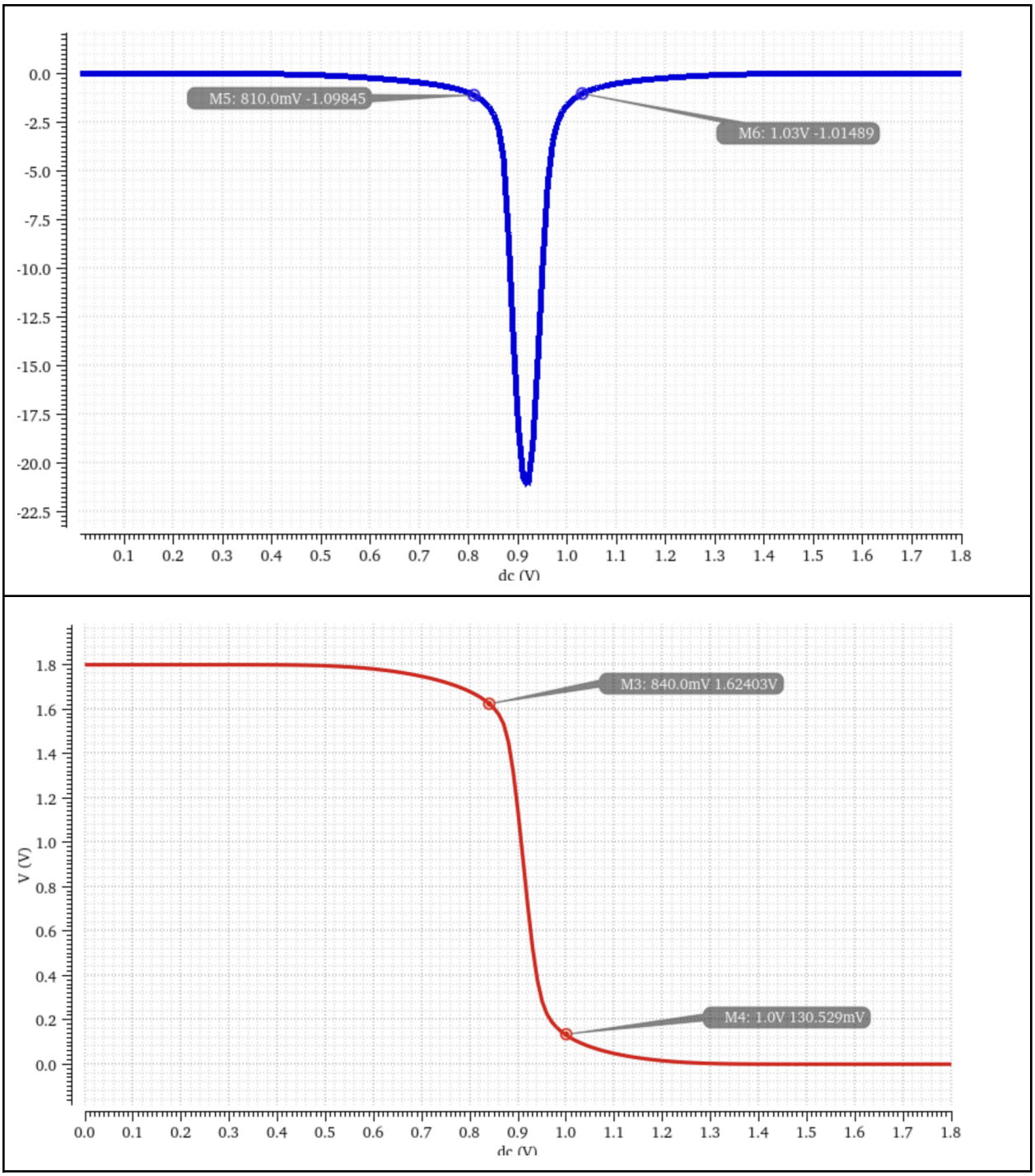


Table 3: Extracted Derivative of CMOS Inverter DC Characteristics.

Output Delay Measurements

Rising Delay (s)	75.16p
Falling Delay (s)	98.11p

Table 4: Output Y Delay Measurements Extracted.

1.6 Conclusion:

This lab examined various characteristics and the layout of a CMOS inverter through Transient, DC, and Parametric analyses. From these experiments, a family of curves was generated to determine the optimal value of *n*, which was found to be 4 and subsequently used for all simulations. The static analysis, based on this value, was utilized to obtain the DC response and calculate VIL,VOH, VIH, VOL, by evaluating the derivative of the voltage transfer curve.

Dynamic simulations were conducted using transient analysis for load capacitances ranging from 0 fF to 150 fF. A physical layout of the CMOS inverter was then created and used to generate the extracted view. The extracted layout exhibited the same static characteristics as the schematic version. Adjusting the PMOS width to twice that of the NMOS did not result in Vm = Vdd/2. Increasing the PMOS width relative to the NMOS was found to both raise Vm and reduce the rise delay.